

# Lecture 10: From Geometric Satake to Geometric Langlands

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# Where We Are

**Lectures 1–9:** Proved the geometric Satake equivalence

$$\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G) \simeq \mathrm{Rep}(\hat{G})$$

as a symmetric monoidal equivalence.

**Recall:**  $LG = G(k((t)))$  (loop group),  $L^+G = G(k[[t]])$  (positive loops),  $\mathrm{Gr}_G = LG/L^+G$  (affine Grassmannian).  $\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G)$  = category of  $L^+G$ -equivariant perverse sheaves on  $\mathrm{Gr}_G$ .  $\hat{G}$  = Langlands dual group (root datum dual to  $G$ ).

**Today:** What does this equivalence **mean**?

- How does it fit into the Langlands program?
- What does it generalize to?
- Where does it lead?

1. **Classical Satake as LLC** — the Satake isomorphism is the unramified case of local Langlands
2. **Geometric Satake and generalizations** — from  $\mathrm{Gr}$  to  $\mathrm{Fl}$ , from Tannakian to categorical
3. **Global Geometric Langlands** — eigensheaves on  $\mathrm{Bun}_G$
4. **Big picture** — the level hierarchy and open problems

# Conventions

- $F$ : non-archimedean local field ( $\mathbb{Q}_p$  or  $\mathbb{F}_q((t))$ ),  $\mathcal{O}_F$ ,  $\mathbb{F}_q$ , uniformizer  $\varpi$
- $W_F$ : Weil group;  $I_F \subset W_F$ : inertia;  $\text{Frob} \in W_F/I_F$ : geometric Frobenius
- ${}^L G = \hat{G} \rtimes \text{Gal}(\bar{F}/F)$ : the  $L$ -group
- $L^+ G = G(\mathcal{O}_F)$ ,  $LG = G(F)$ ,  $\text{Gr}_G = LG/L^+ G$
- $X$ : smooth projective curve over  $\mathbb{F}_q$  (or  $\mathbb{C}$ );  $F_X = \mathbb{F}_q(X)$

# Classical Satake as Local Langlands

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# The $L$ -group

The  $L$ -group of  $G$  sits in a short exact sequence

$$1 \longrightarrow \hat{G} \longrightarrow {}^L G \longrightarrow \mathrm{Gal}(\bar{F}/F) \longrightarrow 1$$

$\hat{G}$  = Langlands dual group (root datum dual to  $G$ ).

Semi-direct product:  ${}^L G = \hat{G} \rtimes \mathrm{Gal}(\bar{F}/F)$ , where  $\sigma \in \mathrm{Gal}$  acts on  $\hat{G}$  via the automorphism induced by  $\sigma$ 's action on  $X^*(T) = X_*(\hat{T})$ .

When  $G$  is **split** (as in this seminar): action is trivial, so  
 ${}^L G = \hat{G} \times \mathrm{Gal}(\bar{F}/F)$ .

## Definition (L-parameter)

A continuous homomorphism

$$\varphi : W_F \times \mathrm{SL}_2(\mathbb{C}) \longrightarrow {}^L G = \hat{G} \rtimes \mathrm{Gal}(\bar{F}/F)$$

satisfying:

1. Projection to  $\mathrm{Gal}(\bar{F}/F)$  agrees with  $W_F \rightarrow \mathrm{Gal}(\bar{F}/F)$
2.  $\varphi|_{W_F}$  is smooth with semisimple image
3.  $\varphi|_{\mathrm{SL}_2}$  is algebraic

Two L-parameters are equivalent if  $\hat{G}$ -conjugate.

# Three Pieces of Data in $\varphi$

The domain  $W_F \times \mathrm{SL}_2(\mathbb{C})$  has three parts:

1. **Frobenius**:  $\varphi(\mathrm{Frob}) \in {}^L G$  — semisimple element = **Satake parameter**  
(when  $\varphi$  is unramified, this is the only data)
2. **Inertia**:  $\varphi|_{I_F} : I_F \rightarrow \hat{G}$  — records **ramification**
3. **Monodromy**:  $\varphi|_{\mathrm{SL}_2} : \mathrm{SL}_2(\mathbb{C}) \rightarrow \hat{G}$  — records **monodromy operator**  
 $N := \log \varphi\left(\begin{smallmatrix} 1 & \\ 0 & 1 \end{smallmatrix}\right) \in \hat{\mathfrak{g}}$  is nilpotent.  $N = 0$  means “no monodromy.”

**Unramified**:  $\varphi|_{I_F} = 1$  and  $N = 0$ , so  $\varphi$  is determined by  $\varphi(\mathrm{Frob})$  alone.

# Centralizer, Component Group, and LLC

## Definition (Centralizer and component group)

For an L-parameter  $\varphi$ :

$$C_\varphi = \{g \in \hat{G} : g\varphi(w)g^{-1} = \varphi(w) \forall w \in W_F \times \mathrm{SL}_2\}$$

$$S_\varphi = \pi_0(C_\varphi / Z(\hat{G})^{\mathrm{Gal}})$$

## Conjecture (Local Langlands Correspondence)

There exists a partition

$$\mathrm{Irr}(G(F)) = \bigsqcup_{\varphi} \Pi_{\varphi}$$

into finite **L-packets**, with injections  $\iota : \Pi_{\varphi} \hookrightarrow \mathrm{Irr}(S_{\varphi})$ .

For  $G = \mathrm{GL}_n$ :  $S_{\varphi} = 1$  always  $\Rightarrow$  L-packets are singletons  $\Rightarrow$  LLC is a bijection.

## Example: $\mathrm{GL}_2$ Centralizer Computation

$G = \mathrm{GL}_2$ ,  $\varphi = \chi_1 \oplus \chi_2$  with  $\chi_1 \neq \chi_2$ :

$$C_\varphi = \{g \in \mathrm{GL}_2(\mathbb{C}) : g \text{ commutes with } \mathrm{diag}(\chi_1(w), \chi_2(w)) \forall w\} = \{\mathrm{diag}(a, b)\}$$

$$S_\varphi = \pi_0((\mathbb{C}^\times)^2 / \mathbb{C}^\times) = 1$$

For  $\chi_1 = \chi_2$ :  $C_\varphi = \mathrm{GL}_2(\mathbb{C})$ , still  $S_\varphi = 1$ .

In fact,  $S_\varphi = 1$  for **all** L-parameters of  $\mathrm{GL}_n$ .

**Theorem** (Harris–Taylor, Henniart; Laumon–Rapoport–Stuhler). *LLC holds for  $\mathrm{GL}_n$  over any non-archimedean local field. Each L-packet is a singleton.*

# The Classical Satake Isomorphism

$K = G(\mathcal{O}_F)$  hyperspecial. Spherical Hecke algebra:

$$\mathcal{H}(G, K) = C_c^\infty(K \backslash G(F) / K).$$

## Theorem (Satake)

$$\mathcal{S} : \mathcal{H}(G, K) \xrightarrow{\sim} \mathbb{C}[X_*(T)]^W \cong R(\hat{G})$$

$$\text{Explicitly: } (Sf)(t) = \delta_B(t)^{1/2} \int_U f(tu) du,$$

$$\delta_B(t) = |\det(\text{Ad}(t)|_{\text{Lie}(U)})|_F.$$

**Example** ( $G = \text{GL}_2$ ,  $F = \mathbb{Q}_p$ ):

$$\mathcal{S} : \mathcal{H}(\text{GL}_2, K) \xrightarrow{\sim} \mathbb{C}[z_1^{\pm 1}, z_2^{\pm 1}]^{S_2}$$

- $\mathcal{S}(1_{K \text{diag}(p,1)K}) = p^{1/2}(z_1 + z_2)$  — character of std
- $\mathcal{S}(1_{K \text{diag}(p,p)K}) = z_1 z_2$  — character of det

# Satake = Unramified LLC

**Proposition.** An unramified L-parameter ( $\varphi|_{I_F} = 1$ ,  $\varphi|_{\mathrm{SL}_2} = 1$ ) is determined by  $\varphi(\mathrm{Frob}) \in \hat{G}^{\mathrm{ss}}/\mathrm{conj}$ .

*Why:*  $W_F/I_F \cong \mathbb{Z}$ , generated by Frob. Since  $\varphi|_{I_F} = 1$ ,  $\varphi$  factors through  $\mathbb{Z}$ , determined by  $\varphi(\mathrm{Frob})$ .

## Corollary (Borel)

$$\{\text{irred. spherical reps of } G(F)\} \cong \{\text{unramified L-parameters}\} / \hat{G}\text{-conj}$$

The bijection:  $\pi \mapsto$  eigenvalue of  $\mathcal{H}(G, K)$  on  $\pi^K$ . Via  $\mathcal{H}(G, K) \cong R(\hat{G})$ , the eigenvalue is a maximal ideal = semisimple conjugacy class =  $\varphi_\pi(\mathrm{Frob})$ .

# Why Satake = Unramified LLC: The Square

$$\begin{array}{ccc}
 \{\text{irred. spherical } \pi\} & \xrightarrow{\sim} & \{\text{unram. } \varphi\} / \hat{G} \\
 \pi \mapsto \pi^K \downarrow & & \downarrow \varphi \mapsto \varphi(\text{Frob}) \\
 \{\text{irred. } \mathcal{H}(G, K)\text{-mod}\} & \xrightarrow[\mathcal{S}^*]{\sim} & \hat{T}/W
 \end{array}$$

**Left:**  $\pi \mapsto \pi^K$  is an equivalence (spherical reps  $\leftrightarrow \mathcal{H}(G, K)$ -modules).

**Bottom:** Satake  $\mathcal{H}(G, K) \xrightarrow{\sim} \mathbb{C}[X_*(T)]^W \cong \mathbb{C}[\hat{T}/W]$ . An irred. module = maximal ideal of  $\mathbb{C}[\hat{T}/W]$  = point of  $\hat{T}/W$ .

**Right:** unramified  $\varphi(\text{Frob}) \in \hat{G}^{\text{ss}}/\text{conj} \cong \hat{T}/W$  (Chevalley restriction).

**Top:** LLC for unramified reps = composite of left, bottom, right<sup>-1</sup>.

# Beyond Satake: Ramified Principal Series

$\chi_1, \chi_2 : \mathbb{Q}_p^\times \rightarrow \mathbb{C}^\times$  with some  $\chi_i$  **ramified** ( $\chi_i|_{\mathbb{Z}_p^\times} \neq 1$ ).

$$\pi(\chi_1, \chi_2) = \text{Ind}_B^{\text{GL}_2}(\chi_1 \otimes \chi_2), \quad \pi(\chi_1, \chi_2)^K = 0$$

L-parameter:  $\varphi = \chi_1 \oplus \chi_2 : W_{\mathbb{Q}_p} \rightarrow \text{GL}_2(\mathbb{C})$

Since  $\chi_i$  ramified:  $\varphi|_{I_{\mathbb{Q}_p}} \neq 1$ , but  $\varphi|_{\text{SL}_2} = 1$ .

**Invisible to  $\mathcal{H}(G, K)$** , but visible to  $\mathcal{H}(G, I)$  when  $\chi_i$  have conductor  $\mathfrak{p}$  (depth zero):  $\pi(\chi_1, \chi_2)$  has a nonzero  $I$ -fixed vector.

For deeper ramification (conductor  $\mathfrak{p}^n$ ,  $n \geq 2$ ): need even smaller compact open  $K_n \subset I$ .

## Beyond Satake: Steinberg (Monodromy Enters)

$\text{St} =$  unique irreducible quotient of  $\text{Ind}_B^{\text{GL}_2}(|\cdot|^{1/2} \otimes |\cdot|^{-1/2})$ .

$\text{St}^K = 0$ , but  $\text{St}' = \mathbb{C}$  (1-dimensional Iwahori-fixed space).

L-parameter:

$$\varphi_{\text{St}}(\text{Frob}) = \begin{pmatrix} p^{1/2} & 0 \\ 0 & p^{-1/2} \end{pmatrix}, \quad \varphi_{\text{St}}|_{\text{SL}_2} = \text{id}$$

$$\varphi_{\text{St}}\left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\right) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad (\text{nilpotent monodromy})$$

The **same**  $\varphi(\text{Frob}) = \text{diag}(p^{1/2}, p^{-1/2})$  with  $\varphi|_{\text{SL}_2} = 1$  gives the *reducible* principal series. The  $\text{SL}_2$  factor distinguishes the irreducible quotient.

# Beyond Satake: Supercuspidal Representations

Cannot be obtained from parabolic induction from any proper Levi.

For  $\mathrm{GL}_2(\mathbb{Q}_p)$ : let  $E/\mathbb{Q}_p$  be quadratic,  $\theta : E^\times \rightarrow \mathbb{C}^\times$  not factoring through  $\mathrm{Nm}_{E/F}$ .

$$\pi_\theta = \mathrm{c}\text{-Ind}_{E^\times}^{\mathrm{GL}_2(\mathbb{Q}_p)} \theta \quad (\text{supercuspidal})$$

L-parameter: **irreducible** 2-dimensional representation

$$\varphi_\theta = \mathrm{Ind}_{W_E}^{W_{\mathbb{Q}_p}} \theta : W_{\mathbb{Q}_p} \rightarrow \mathrm{GL}_2(\mathbb{C})$$

$\varphi_\theta|_{\mathrm{SL}_2} = 1$  but  $\varphi_\theta|_{I_{\mathbb{Q}_p}}$  nontrivial and  $\varphi_\theta$  irreducible.

$\pi_\theta^K = 0$ ,  $\pi_\theta^I = 0$  (for deep ramification) — invisible to both  $\mathcal{H}(G, K)$  and  $\mathcal{H}(G, I)$ .

# Summary Table for $GL_2$

| $\pi$             | $\varphi(\text{Frob})$           | $\varphi _{I_F}$ | $\varphi _{\text{SL}_2}$ | Visible to          |
|-------------------|----------------------------------|------------------|--------------------------|---------------------|
| Unram. p.s.       | $\text{diag}(\alpha, \beta)$     | triv             | triv                     | $\mathcal{H}(G, K)$ |
| Depth-0 ram. p.s. | $\chi_1 \oplus \chi_2$           | ntriv            | triv                     | $\mathcal{H}(G, I)$ |
| Steinberg         | $\text{diag}(p^{1/2}, p^{-1/2})$ | triv             | id                       | $\mathcal{H}(G, I)$ |
| Supercusp. (d.0)  | $\text{Ind}_{W_E}^{W_F} \theta$  | ntriv            | triv                     | $\mathcal{H}(G, I)$ |
| Supercusp. (deep) | $\text{Ind}_{W_E}^{W_F} \theta$  | wild             | triv                     | deeper level        |

## Beyond $\mathrm{GL}_n$ : Nontrivial L-packets

For  $G = \mathrm{SL}_2$ ,  $\hat{G} = \mathrm{PGL}_2(\mathbb{C})$ . Consider  $\varphi : W_F \rightarrow \mathrm{PGL}_2(\mathbb{C})$  lifting to  $\tilde{\varphi}(\mathrm{Frob}) = \mathrm{diag}(\alpha, -\alpha)$ .

$$C_\varphi = \mathrm{Cent}_{\mathrm{PGL}_2}(\mathrm{Im} \varphi) \cong \mathbb{Z}/2\mathbb{Z}, \quad S_\varphi = \mathbb{Z}/2\mathbb{Z}$$

$\Rightarrow$  L-packet  $\Pi_\varphi$  contains **two** representations of  $\mathrm{SL}_2(F)$ .

### Three axes of complexity in LLC:

1. **Ramification** ( $\varphi|_{I_F}$ ): trivial  $\rightarrow$  tame  $\rightarrow$  wild
2. **Monodromy** ( $\varphi|_{\mathrm{SL}_2}$ ): trivial  $\rightarrow$  nontrivial
3. **L-packet structure** ( $S_\varphi$ ): trivial ( $\mathrm{GL}_n$ )  $\rightarrow$  nontrivial (general  $G$ )

Satake sits at the **origin** of all three axes.

# From Classical to Categorical: The Full Circle



For Satake:

- Classical:  $\mathcal{H}_G \cong R(\hat{G})$
- Categorical:  $\text{Perv}_{L+G}(\text{Gr}_G) \simeq \text{Rep}(G^\vee)$
- Decategorification:  $\text{Rep}(\hat{G}) \xrightarrow{\text{Tr}_\sigma} \text{Coh}_{\text{fr}}^{\hat{G}}(\hat{G}\sigma) \xrightarrow{\text{End}(\tilde{\mathbf{I}})} R(\hat{G}) \cong \mathcal{H}_G$

For the full LLC, each axis corresponds to a geometric refinement:

- **Ramification**  $\leftrightarrow$  relaxing equivariance ( $L^+G \rightarrow I \rightarrow \text{smaller}$ )
- **Monodromy**  $\leftrightarrow$   $\text{SL}_2$ -factor in categorical trace formalism
- **L-packet structure**  $\leftrightarrow$  categorical structure (IndCoh vs. QCoh)

# **Geometric Satake and Generalizations**

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## Review: The Affine Grassmannian

$G$  connected reductive over  $k$ ,  $F = k((t))$ ,  $\mathcal{O} = k[[t]]$ .

$$\mathrm{Gr}_G = LG/L^+G \quad (\text{ind-projective ind-scheme})$$

For  $\mathrm{GL}_n$ : an  $R$ -point of  $\mathrm{Gr}_{\mathrm{GL}_n}$  is an  $R$ -family of **lattices**  $\Lambda \subset R((t))^n$  with  $\Lambda \otimes_{R[[t]]} R((t)) = R((t))^n$ .

Standard lattice:  $\Lambda_0 = R[[t]]^n$ .

## $GL_2$ : Schubert Cells on $Gr$

$T$  = diagonal torus,  $B$  = upper triangular. Dominant coweights:  
 $\lambda = (a, b)$ ,  $a \geq b$ .

| $\lambda$ | $\dim Gr^\lambda$ | Description                                                                                                         |
|-----------|-------------------|---------------------------------------------------------------------------------------------------------------------|
| $(0, 0)$  | 0                 | $\{\Lambda_0\}$ (standard lattice)                                                                                  |
| $(1, 0)$  | 1                 | $\{\Lambda : \Lambda_0 \subset \Lambda \subset t^{-1}\Lambda_0, \dim_k(\Lambda/\Lambda_0) = 1\} \cong \mathbb{P}^1$ |
| $(1, 1)$  | 0                 | $\{t^{-1}\Lambda_0\}$ (scalar shift)                                                                                |
| $(2, 0)$  | 2                 | $\Lambda_0 \subset \Lambda \subset t^{-2}\Lambda_0, \dim_k(\Lambda/\Lambda_0) = 2$                                  |

Cartan decomposition:  $LG = \bigsqcup_{\lambda \in X_*(T)^+} L^+G \cdot t^\lambda \cdot L^+G$ .

**Dimension formula** (Lecture 4):  $\dim Gr^\lambda = \langle 2\rho, \lambda \rangle$  where  
 $\rho = \frac{1}{2} \sum_{\alpha > 0} \alpha$ .

For  $GL_2$ :  $2\rho = (1, -1)$ , so  $\dim Gr^{(a,b)} = a - b$ .

# The Affine Flag Variety

Recall:  $t$  is a uniformizer of  $F = k((t))$ , so  $\mathcal{O} = k[[t]]$ ,  $L^+G = G(k[[t]])$ ,  $LG = G(k((t)))$ .

$\mathrm{Gr}_G = LG/L^+G$  uses **hyperspecial**  $L^+G$ , captures only unramified reps.

Replace by **Iwahori**  $I = \mathrm{ev}_0^{-1}(B) \subset L^+G$  (preimage of Borel at  $t = 0$ ).

## Definition

The *affine flag variety* is  $\mathrm{Fl}_G = LG/I$ , with  $\pi : \mathrm{Fl}_G \rightarrow \mathrm{Gr}_G$  (fiber  $L^+G/I \cong G/B$ ).

**Modular interpretation** ( $\mathrm{GL}_n$ ):  $\mathrm{Fl}_{\mathrm{GL}_n}$  parametrizes **lattice chains**:

$$\mathrm{Fl}_{\mathrm{GL}_n}(R) = \left\{ (\Lambda_0 \supset \Lambda_1 \supset \cdots \supset \Lambda_{n-1} \supset t\Lambda_0) \left| \begin{array}{l} \text{each } \Lambda_i \text{ a lattice} \\ \Lambda_i/\Lambda_{i+1} \text{ loc. free rk } 1 \end{array} \right. \right\}$$

$\pi : \mathrm{Fl}_{\mathrm{GL}_n} \rightarrow \mathrm{Gr}_{\mathrm{GL}_n}$  sends the chain to  $\Lambda_0$ .

Iwahori subgroup:

$$I = \left\{ \begin{pmatrix} a(t) & b(t) \\ c(t) & d(t) \end{pmatrix} \in \mathrm{GL}_2(k[[t]]) \mid c(0) = 0 \right\}$$

A  $k$ -point of  $\mathrm{Fl}_{\mathrm{GL}_2}$ : a pair  $(\Lambda, \ell)$  where  $\Lambda$  is a lattice and  $\ell \subset \Lambda/t\Lambda \cong k^2$  is a line (= complete flag in 2d fiber).

$\pi : \mathrm{Fl}_{\mathrm{GL}_2} \rightarrow \mathrm{Gr}_{\mathrm{GL}_2}$  forgets  $\ell$ . Fiber =  $\mathbb{P}(\Lambda/t\Lambda) \cong \mathbb{P}^1$ .

$\mathrm{Gr}_G$ : Schubert cells indexed by  $X_*(T)^+ = \{(a, b) : a \geq b\}$

$\mathrm{Fl}_G$ : cells indexed by  $\widetilde{W}_{\mathrm{aff}} = X_*(T) \rtimes W$  (**no folding**)

Over  $\mathrm{Gr}^{(1,0)} \cong \mathbb{P}^1$ : the flag variety refines into **two**  $I$ -orbits.

# Why the Iwahori Step Is Fundamental

$\mathcal{H}(G, I) = C_c^\infty(I \backslash G(F) / I)$  occupies a privileged position:

1. **Generators and relations.**  $T_s$  for simple affine reflections  $s \in W_{\text{aff}}$ :

$$(T_s - q)(T_s + 1) = 0 \quad + \text{ braid relations}$$

This is the **Iwahori–Hecke algebra** of Kazhdan–Lusztig theory.

2. **Bernstein’s theorem.**  $Z(\mathcal{H}(G, I)) \cong \mathbb{C}[X_*(T)]^W \cong \mathcal{H}(G, K)$   
Satake  $\hookrightarrow$  Iwahori–Hecke **as the center**.
3. **Coverage.**  $I$ -fixed vectors: unramified, depth-zero ramified p.s., Steinberg, depth-zero supercuspidals.
4. **Beyond Iwahori.**  $\mathcal{H}(G, K_n)$  (Moy–Prasad) has no KL basis, no quadratic relation  $\Rightarrow$  wild ramification largely open.

$\mathcal{H}(G, I)$  is **not commutative** (unlike  $\mathcal{H}(G, K)$ ).

## Review: IC Sheaves on $\mathrm{Gr}_G$

The IC sheaf on the Schubert variety  $\overline{\mathrm{Gr}}^\lambda$ :

$$\mathrm{IC}_\lambda = j_{!*}(\underline{k}_{\mathrm{Gr}^\lambda}[\langle 2\rho, \lambda \rangle])$$

$j : \mathrm{Gr}^\lambda \hookrightarrow \overline{\mathrm{Gr}}^\lambda$  open,  $j_{!*}$  = intermediate extension.

Characterized by: (1)  $\mathrm{IC}_\lambda|_{\mathrm{Gr}^\lambda} = \underline{k}[\langle 2\rho, \lambda \rangle]$ , (2) support condition, (3) cosupport condition.

**Example** ( $\mathrm{GL}_2$ ):

- $\mathrm{IC}_{(0,0)} = k_{\{L_0\}}$  (skyscraper at origin)
- $\mathrm{IC}_{(1,0)} = \underline{k}_{\mathbb{P}^1}[1]$  (constant sheaf on  $\mathbb{P}^1$ , shifted)
- $\mathrm{IC}_{(1,1)} = k_{\{t^{-1}\Lambda_0\}}$  (skyscraper)
- $\mathrm{IC}_{(2,0)}$ : IC of  $\overline{\mathrm{Gr}}^{(2,0)}$ , *not* shifted constant (singular!)

Simple objects of  $\mathrm{Perv}_{L+G}(\mathrm{Gr}_G) = \{\mathrm{IC}_\lambda\}_{\lambda \in X_*(T)^+}$ .

# Review: Convolution Product

(Lecture 6; [BR17, §4–5])

$$\mathcal{F}_1 \star \mathcal{F}_2 = m_! ((q^*)^{-1} p^* (\mathcal{F}_1 \boxtimes \mathcal{F}_2))$$

via the convolution diagram:

$$\mathrm{Gr}_G \times \mathrm{Gr}_G \xleftarrow{p} LG \times \mathrm{Gr}_G \xrightarrow{q} LG \times^{L^+G} \mathrm{Gr}_G \xrightarrow{m} \mathrm{Gr}_G$$

**Example** ( $\mathrm{GL}_2$ ):

$$\mathrm{IC}_{(1,0)} \star \mathrm{IC}_{(1,0)} = \mathrm{IC}_{(2,0)} \oplus \mathrm{IC}_{(1,1)}$$

corresponding to  $\mathrm{std} \otimes \mathrm{std} = \mathrm{Sym}^2 \oplus \det$  in  $\mathrm{Rep}(\mathrm{GL}_2^\vee)$ . ✓

# IC Sheaves and Convolution on $\mathrm{Fl}_G$

**Generalization:** on  $\mathrm{Fl}_G$ , simple objects of  $\mathrm{Perv}_I(\mathrm{Fl}_G)$  are  $\mathrm{IC}_w$ , indexed by  $w \in W_{\mathrm{aff}}$  — **infinitely many** more than on  $\mathrm{Gr}_G$ .

Convolution  $\star$  replaces  $L^+G$  by  $I$ : twisted product  $LG \times^I \mathrm{Fl}_G$ .

**Crucial failure:**  $\mathrm{Perv}_I(\mathrm{Fl}_G)$  is **not semisimple!**

$\exists$  nontrivial  $\mathrm{Ext}^1(\mathrm{IC}_w, \mathrm{IC}_{w'}) \neq 0$ .

Parity vanishing (Lecture 7) relied on  $\mathrm{Gr}^\lambda$  being an affine bundle over a partial flag variety — this breaks for Iwahori orbits.

**Rep-theoretic counterpart:** Steinberg sits in a nonsplit exact sequence

$$0 \rightarrow \mathbf{1} \rightarrow \mathrm{Ind}_B^G(| \cdot |^{1/2} \otimes | \cdot |^{-1/2}) \rightarrow \mathrm{St} \rightarrow 0$$

# Review: Fiber Functor and Tannakian Reconstruction

(Lectures 7–8; [BR17, Theorem 5.9])

The fiber functor  $F = \bigoplus_{\mu} F_{\mu} : \text{Perv}_{L+G}(\text{Gr}_G) \rightarrow \text{Vect}_k$ :

$$F_{\mu}(\mathcal{A}) = H_{T_{\mu}}^{\langle 2\rho, \mu \rangle}(\text{Gr}_G, \mathcal{A}) \cong H_c^{\langle 2\rho, \mu \rangle}(S_{\mu}, \mathcal{A})$$

$S_{\mu} = N((t)) \cdot t^{\mu}$  (semi-infinite orbit). Parity vanishing  $\Rightarrow$  direct sum decomposition.

**Example** ( $\text{GL}_2$ ):  $\text{IC}_{(1,0)} \leftrightarrow \text{std}$ : weights  $(1, 0), (0, 1)$ , each  $F_{\mu} = k$ .

$\text{IC}_{(2,0)} \leftrightarrow \text{Sym}^2$ : weights  $(2, 0), (1, 1), (0, 2)$ , each multiplicity 1.  $\checkmark$

Tannakian reconstruction (Lecture 8; [DM82]):

$$\text{Perv}_{L+G}(\text{Gr}_G, k) \simeq \text{Rep}_k(G_k^{\vee})$$

# What Replaces $\text{Rep}(G^\vee)$ at Deeper Levels?

Tannakian formalism requires: (i) semisimple abelian category, (ii) fiber functor to  $\text{Vect}$ .

At Iwahori level:  $\text{Perv}_I(\text{Fl}_G)$  fails (i)  $\Rightarrow$  Tannakian does not apply.

Instead: work at the **derived/DG categorical** level.

| Level                 | Automorphic                                                 | Spectral                                   |
|-----------------------|-------------------------------------------------------------|--------------------------------------------|
| $L^+G$ (hyperspecial) | $\text{Perv}_{L^+G}(\text{Gr}_G) \simeq \text{Rep}(G^\vee)$ | Tannakian                                  |
| $I$ (Iwahori)         | $D_I(\text{Fl}_G)$                                          | $\text{Coh}(\hat{\mathfrak{g}}^*/\hat{G})$ |
| Deeper                | $D_{K_n}(\text{Fl}_G)$                                      | ? (open)                                   |

At Iwahori: spectral side involves coherent sheaves on dual Lie algebra or Steinberg variety — accommodates extensions and monodromy [AB09].

# Classical Motivation for the Trace

For an algebra  $A$  over  $E$ :

$$\mathrm{Tr}(A) = A/[A, A]$$

Universal trace function: any  $f : A \rightarrow V$  with  $f(ab) = f(ba)$  factors through  $\mathrm{Tr}$ .

Equivalently:  $\mathrm{Tr}(A) = \text{coequalizer of } A \otimes A \rightrightarrows A$  given by  $a \otimes b \mapsto ab$  and  $a \otimes b \mapsto ba$ .

# Categorical Trace: Definition

Let  $(\mathcal{C}, \otimes, \mathbf{1})$  be an  $E$ -linear monoidal category,  $\sigma : \mathcal{C} \rightarrow \mathcal{C}$  a monoidal endofunctor.

## Definition (Categorical trace; [Zhu18, §4]; Lecture 9)

$\mathrm{Tr}_\sigma(\mathcal{C})$  is the 2-colimit:

$$\mathcal{C}^{\otimes 3} \begin{matrix} \rightarrow \\ \rightarrow \\ \rightarrow \end{matrix} \mathcal{C}^{\otimes 2} \rightrightarrows \mathcal{C}$$

Two functors  $\mathcal{C}^{\otimes 2} \rightarrow \mathcal{C}$ :  $(X, Y) \mapsto X \otimes Y$  and  $(X, Y) \mapsto Y \otimes \sigma X$ .

**Concretely:** Objects of  $\mathrm{Tr}_\sigma(\mathcal{C}) =$  objects of  $\mathcal{C}$  (denoted  $\tilde{X}$ ). Morphisms:

$$\mathrm{Hom}_{\mathrm{Tr}_\sigma}(\tilde{X}, \tilde{Y}) = \left( \bigoplus_{(V, W)} \mathrm{Hom}(X, V \otimes W) \otimes_E \mathrm{Hom}(W \otimes \sigma V, Y) \right) / \sim$$

**Key:** in  $\mathrm{Tr}_\sigma(\mathcal{C})$ ,  $\widetilde{X \otimes Y} \cong \widetilde{Y \otimes \sigma X}$  — enforces “ $\sigma$ -twisted commutativity.”

## The Key Computation: $\mathrm{Tr}_\sigma(\mathrm{Rep}(\hat{G}))$

$\hat{G}$  Langlands dual,  $\sigma : \hat{G} \rightarrow \hat{G}$   $q$ -Frobenius (for split  $G$ :  $\sigma = \mathrm{id}$ ).

**$\sigma$ -twisted conjugation:**  $c_\sigma(g) \cdot h = g^{-1}h\sigma(g)$ ,  $g \in \hat{G}$ ,  $h \in \hat{G}\sigma$ .

$\mathrm{Coh}^{\hat{G}}(\hat{G}\sigma) = \hat{G}$ -equivariant coherent sheaves on  $[\hat{G}\sigma/\hat{G}]$ .

For  $V \in \mathrm{Rep}(\hat{G})$ : define  $\tilde{V} \in \mathrm{Coh}^{\hat{G}}(\hat{G}\sigma)$  with global sections  $\mathcal{O}_{\hat{G}} \otimes V$ ,  $\hat{G}$ -action  $g \cdot f(h) = g(f(g^{-1}h\sigma(g)))$ .

$\mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma) =$  full subcategory spanned by such  $\tilde{V}$ .

**Proposition ([Lee26, Prop. 2.4]; [Zhu18])**

$\mathrm{Tr}_\sigma(\mathrm{Rep}(\hat{G})) \cong \mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma)$ , sending  $\tilde{V} \mapsto \tilde{V}$ .

$\mathrm{Hom}_{\mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}}(\tilde{V}_1, \tilde{V}_2) = (\mathcal{O}_{\hat{G}} \otimes V_1^* \otimes V_2)^{\hat{G}}$ .

# Local Shtukas and the Decategorification Theorem

Define the moduli of **local Shtukas**:

$$\mathrm{Sht}^{\mathrm{loc}} = LG / \mathrm{Ad}_\sigma L^+ G$$

( $LG$  modulo  $\sigma$ -twisted conjugation  $g \mapsto h^{-1}g\sigma(h)$  by  $h \in L^+ G$ ).

**Notation guide:**

- $\mathrm{Hk}^{\mathrm{loc}} = \mathrm{Gr}_G \times_{\mathrm{Gr}_G} \mathrm{Gr}_G$ : local Hecke stack (parametrizes pairs of modifications)
- $\mathrm{P}(\mathrm{Hk}^{\mathrm{loc}})$ : perverse sheaves on  $\mathrm{Hk}^{\mathrm{loc}}$ ; equiv. to  $\mathrm{Perv}_{L^+ G}(\mathrm{Gr}_G)$  via Satake
- $\Phi^{\mathrm{loc}}$ : the *local Frobenius* map  $\mathrm{Hk}^{\mathrm{loc}} \rightarrow \mathrm{Sht}^{\mathrm{loc}}$  (sends a modification to its Frobenius twist)
- $\mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})$ : perverse sheaves on  $\mathrm{Sht}^{\mathrm{loc}}$  equipped with Hecke structure (= action of  $\mathrm{Rep}(\hat{G})$  via cohomological correspondences)

$\delta_1 \in \mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})$ : image of  $\mathrm{IC}_0 \in \mathrm{Perv}_{L^+ G}(\mathrm{Gr})$  under

$$\mathrm{Perv}_{L^+ G}(\mathrm{Gr}) \xrightarrow{\sim} \mathrm{P}(\mathrm{Hk}^{\mathrm{loc}}) \xrightarrow{\Phi^{\mathrm{loc}}} \mathrm{P}(\mathrm{Sht}^{\mathrm{loc}}).$$

**Theorem ([Zhu18, Thm 4.3.1]; [Lee26, Thm 4.9])**

(1)  $\exists$  fully faithful  $S : \mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma) \rightarrow \mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})$ .

(2)  $S(\mathcal{O}_{[\hat{G}\sigma/\hat{G}]}) = \delta_1$ , and on endomorphisms:

$$S : \overline{\mathbb{Q}}_\ell[\hat{G}\sigma]^{\hat{G}} = \mathrm{End}(\tilde{\mathbf{1}}) \longrightarrow \mathrm{End}(\delta_1) \cong \mathcal{H}_G \otimes \overline{\mathbb{Q}}_\ell$$

is the **classical Satake isomorphism**.

# The Commutative Diagram

$$\begin{array}{ccc}
 \mathrm{Rep}(\hat{G}) & \xrightarrow[\sim]{\mathrm{Sat}} & \mathrm{P}(\mathrm{Hk}^{\mathrm{loc}}) \\
 \downarrow V \mapsto \tilde{V} & & \downarrow \Phi^{\mathrm{loc}} \\
 \mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma) & \xrightarrow{S} & \mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})
 \end{array}$$

**Top row:** categorical Geometric Satake (equivalence of categories)

**Left column:** categorical trace ( $V \mapsto \tilde{V}$ )

**Decategorification:** take  $\mathrm{End}(\tilde{\mathbf{1}})$ :

$$\underbrace{\mathrm{Rep}(\hat{G})}_{\text{category}} \xrightarrow{\mathrm{Tr}_{\sigma}} \underbrace{\mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma)}_{\text{category}} \xrightarrow{\mathrm{End}(\tilde{\mathbf{1}})} \underbrace{R(\hat{G}) \cong \mathcal{H}_G}_{\text{algebra}}$$

For split  $G$ :  $\mathrm{End}(\tilde{\mathbf{1}}) = \overline{\mathbb{Q}}_{\ell}[\hat{G}]^{\hat{G}} = R(\hat{G}) \otimes \overline{\mathbb{Q}}_{\ell}$ .

# Decategorification at Each Level

## Theorem (Zhu [Zhu25, Thm 1.6])

The categorical trace extends to Iwahori/tame level. The  $\sigma$ -twisted trace of the Iwahori-level equivalence recovers the tame LLC.

| Categorical (geom. side)                                   | $\xrightarrow{\text{Tr}_\sigma}$ | Classical LLC                                       |
|------------------------------------------------------------|----------------------------------|-----------------------------------------------------|
| $\text{Perv}_{L+G}(\text{Gr}_G) \simeq \text{Rep}(G^\vee)$ | $\rightarrow$                    | $\mathcal{H}(G, K) \cong R(\hat{G})$                |
| $D_I(\text{Fl}_G) \simeq (\text{spectral})$                | $\rightarrow$                    | $\mathcal{H}(G, I) \leftrightarrow \text{tame LLC}$ |
| $D_{K_n}(?) \simeq ?$                                      | $\rightarrow$                    | wild LLC ( <b>open</b> )                            |

**Open Problem.** Extend categorical LLC to wild ramification. Difficulty: wild inertia destroys semisimplicity at all levels.

# Global Geometric Langlands

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# The Local–Global Dictionary

| Local (at $x \in X$ )                                             | Global (on curve $X$ )                                               |
|-------------------------------------------------------------------|----------------------------------------------------------------------|
| Local field $F_x = k((t))$                                        | Function field $F_X = k(X)$                                          |
| $\mathrm{Gr}_G = LG/L^+G$                                         | $\mathrm{Bun}_G(X)$                                                  |
| L-parameter $\varphi_x : W_{F_x} \rightarrow {}^L G$              | $\sigma : \pi_1(X) \rightarrow G^\vee$                               |
| $\mathrm{Perv}_{L^+G}(\mathrm{Gr}_G) \simeq \mathrm{Rep}(G^\vee)$ | $D(\mathrm{Bun}_G) \simeq \mathrm{IndCoh}(\mathrm{LocSys}_{G^\vee})$ |

**Local-global compatibility:**  $\sigma$  on  $X$  restricts at each  $x$  to  $\sigma|_{W_{F_x}}$ .

If  $\sigma$  unramified at  $x$ :  $\sigma|_{W_{F_x}}$  is unramified  $\Rightarrow$  Satake governs.

At finitely many “bad” points: need deeper level structures from §3.

$X$  smooth projective curve over  $\mathbb{F}_q$  (or  $\mathbb{C}$ ).

$$\mathrm{Bun}_G(X) = \{\text{principal } G\text{-bundles on } X\} \quad (\text{algebraic stack})$$

- Smooth, dimension  $(g-1)\dim G$
- $\mathrm{Bun}_{\mathrm{GL}_n}$  = moduli of rank- $n$  vector bundles
- $\pi_0(\mathrm{Bun}_G) \cong \pi_1(G)$

$\mathrm{Bun}_G$  is the **geometric analog** of  $G(F) \backslash G(\mathbb{A}_F) / \prod_v K_v$ .

**Example** ( $\mathrm{GL}_2$  on  $\mathbb{P}^1$ ): by Grothendieck,  $\mathcal{E} \cong \mathcal{O}(a) \oplus \mathcal{O}(b)$ ,  $a \geq b$ .

$$\mathrm{LocSys}_{G^\vee}(X) = \{\pi_1^{\mathrm{\acute{e}t}}(X) \rightarrow G^\vee(\overline{\mathbb{Q}_\ell})\} / G^\vee\text{-conj}$$

## Definition (Hecke stack at $x$ )

$$\mathrm{Hk}_x = \left\{ (\mathcal{E}, \mathcal{E}', \alpha) \mid \mathcal{E}, \mathcal{E}' \in \mathrm{Bun}_G, \alpha : \mathcal{E}|_{X \setminus \{x\}} \xrightarrow{\sim} \mathcal{E}'|_{X \setminus \{x\}} \right\}$$

By Beauville–Laszlo [BL95]: fibers of  $\mathrm{Hk}_x \rightarrow \mathrm{Bun}_G$  are  $\cong \mathrm{Gr}_G$ .

**Example** ( $\mathrm{GL}_2$  on  $\mathbb{P}^1$ ):  $\mathcal{E} = \mathcal{O} \oplus \mathcal{O}$ . A Hecke modification of type  $(1, 0)$  at  $x = 0$ : choose a line  $\ell \subset \mathcal{E}_0 \cong k^2$ , giving  $\mathcal{E}' = \mathcal{O}(1) \oplus \mathcal{O}(-1)$ . The modification is parametrized by  $\mathbb{P}^1$  (the choice of  $\ell$ ).

# Hecke Operators from Geometric Satake

For  $V \in \text{Rep}(G^\vee)$ , the Geometric Satake correspondence  $\text{IC}_V \in \text{Perv}_{L+G}(\text{Gr}_G)$  defines:

## Definition (Hecke functor)

Via the Hecke correspondence

$$\text{Bun}_G \xleftarrow{\text{pr}_1} \text{Hk}_G \xrightarrow{\text{pr}_2} \text{Bun}_G$$

where  $\text{pr}_1(\mathcal{E}, \mathcal{E}', x, \alpha) = \mathcal{E}$  and  $\text{pr}_2(\mathcal{E}, \mathcal{E}', x, \alpha) = \mathcal{E}'$ :

$$T_V : D(\text{Bun}_G) \rightarrow D(\text{Bun}_G), \quad T_V(\mathcal{F}) = (\text{pr}_2)_! (\text{pr}_1^*(\mathcal{F}) \otimes \text{IC}_V)$$

Monoidal structure:  $T_V \circ T_W \simeq T_{V \otimes W}$ .

**Geometric Satake tells us which  $T_V$  to consider** — one for each representation of  $\hat{G}$ .

# Hecke Eigensheaves

Given  $\sigma : \pi_1(X) \rightarrow G^\vee$  (a  $G^\vee$ -local system on  $X$ ) and  $V \in \text{Rep}(G^\vee)$ , define  $V_\sigma$ : the **local system on  $X$**  obtained by composing  $\sigma$  with the representation  $G^\vee \rightarrow \text{GL}(V)$ . Concretely,  $V_\sigma$  is the vector bundle with flat connection on  $X$  whose monodromy is  $\sigma$  acting through  $V$ .

## Definition

$\mathcal{F} \in D(\text{Bun}_G)$  is a **Hecke eigensheaf** with eigenvalue  $\sigma \in \text{LocSys}_{G^\vee}(X)$  if for all  $V \in \text{Rep}(G^\vee)$ :

$$T_V(\mathcal{F}) \simeq V_\sigma \boxtimes \mathcal{F}$$

**Analogy:**  $T_p f = \lambda_p f$  for classical Hecke eigenforms.

Here the “eigenvalue” is not a number but a **local system on  $X$** .

Function-sheaf dictionary:  $\mathcal{F} \mapsto f_{\mathcal{F}}$  (automorphic form) with  $T_V(f_{\mathcal{F}}) = \text{Tr}(\text{Frob}_x, V_\sigma) \cdot f_{\mathcal{F}}$ .

# Categorical Framework: $D\text{-mod}$ , $QCoh$ , $IndCoh$

The Geometric Langlands conjecture is an equivalence of **DG categories**. The key players:

$D\text{-mod}(Y)$  = derived category of  $D$ -modules on a stack  $Y$  (= “sheaves with flat connection”). For  $Y = \text{Bun}_G$ , this is the automorphic side.

$QCoh(Y)$  = derived category of quasi-coherent sheaves on  $Y$ . When  $Y$  is smooth,  $QCoh(Y)$  and  $Coh(Y)$  are closely related.

$IndCoh(Y)$  = ind-completion of  $Coh(Y)$  (coherent sheaves on  $Y$ ).  
Agrees with  $QCoh(Y)$  when  $Y$  is smooth, but differs when  $Y$  is singular.  
 $LocSys_{\hat{G}}$  is typically **singular** (e.g., reducible local systems), so  $IndCoh$  is essential — it remembers the singularity structure that  $QCoh$  forgets.

# The Geometric Langlands Conjecture

## Conjecture (Beilinson–Drinfeld, Arinkin–Gaitsgory [AG15])

$$\mathrm{D}\text{-mod}(\mathrm{Bun}_G) \simeq \mathrm{IndCoh}(\mathrm{LocSys}_{\hat{G}})$$

(equivalence of DG categories)

- **Automorphic side:**  $\mathrm{D}\text{-mod}(\mathrm{Bun}_G)$  — D-modules on the moduli of  $G$ -bundles
- **Spectral side:**  $\mathrm{IndCoh}(\mathrm{LocSys}_{\hat{G}})$  — ind-coherent sheaves on the stack of  $\hat{G}$ -local systems
- Hecke eigensheaves  $\leftrightarrow$  skyscraper sheaves on  $\mathrm{LocSys}_{\hat{G}}$

## Theorem (Gaitsgory–Raskin–et al. [GR+24], 2024)

This holds in the de Rham setting  $(X/\mathbb{C}, \text{any reductive } G)$ .

# Global with Level Structure

Without level: Hecke operators use  $IC_V$  on  $Gr_G$   
(hyperspecial/unramified).

To incorporate ramification: introduce **level structures** at finitely many points.

| Local input at $x$ | Global moduli | Hecke at $x$                                 |
|--------------------|---------------|----------------------------------------------|
| $Gr_G = LG/L^+G$   | $Bun_G$       | $T_V$ from $IC_V \in \text{Perv}_{L^+G}(Gr)$ |
| $Fl_G = LG/I$      | $Bun'_G$      | Hecke from $\text{Perv}_I(Fl_G)$             |

Iwahori level at  $x$ : replace  $Bun_G$  by  $G$ -bundles with a **Borel reduction** of the fiber  $\mathcal{E}_x$ , and use IC sheaves on  $Fl_G$  at  $x$ .

# How Geometric Satake Enters the Global Picture

Geometric Satake is the **local input** for the global correspondence:

1. **Defines the dual group**  $\hat{G}$  (Tannakian reconstruction from  $\text{Perv}_{L+G}(\text{Gr}_G)$ )
2. **Provides Hecke operators** ( $\text{IC}_V$  defines  $T_V$ )
3. **Starting point** of the level hierarchy (hyperspecial  $\rightarrow$  Iwahori  $\rightarrow \dots$ )

Without geometric Satake, we would not know:

- What  $\hat{G}$  is (at the categorical level)
- Which Hecke operators to consider
- What “eigenvalue = local system” means

# The $p$ -adic Analog and Big Picture

---

# Fargues–Scholze: $p$ -adic Geometric Langlands

Replace the function field curve  $X/\mathbb{F}_q$  by the **Fargues–Fontaine curve**  $X_{\text{FF}}$ : a “curve” over  $\mathbb{Q}_p$  whose geometry encodes  $p$ -adic Hodge theory.

## Theorem (Fargues–Scholze [FS21, Thm I.4.1])

$\text{Bun}_G(X_{\text{FF}})$  has strata indexed by  $B(G)$  (Kottwitz set = isocrystal classes), with each stratum  $\text{Bun}_G^b$  satisfying  $D(\text{Bun}_G^b) \simeq D(G_b(E))$ , where  $G_b$  is the inner form of  $G$  associated to  $b \in B(G)$ .

## Conjecture (Fargues–Scholze, Zhu, Hansen–Mann)

$$D(\text{Bun}_G(X_{\text{FF}})) \simeq \text{IndCoh}(\text{Par}_{G^\vee})$$

where  $\text{Par}_{G^\vee}$  = stack of  $L$ -parameters  $W_E \rightarrow {}^L G$  (the “parameter space”).

KIAS Categorical Langlands Summit (June 14–26, 2026, Seoul):  
Hansen–Mann ( $\text{GL}_n$ ), Zhu (tame case).

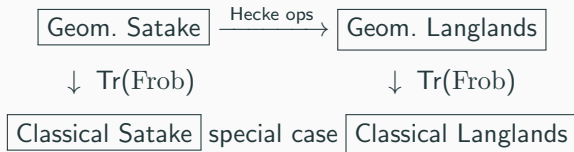
# The Full Landscape

|                   | Automorphic                                             | Spectral                                              |
|-------------------|---------------------------------------------------------|-------------------------------------------------------|
| Classical local   | $\mathcal{H}(G, K)$                                     | $\mathbb{C}[\hat{T}/W]$                               |
| Geometric local   | $\mathrm{Perv}_{L+G}(\mathrm{Gr}_G)$                    | $\mathrm{Rep}(\hat{G})$                               |
| Categorical local | $\mathrm{P}^{\mathrm{Hk}}(\mathrm{Sht}^{\mathrm{loc}})$ | $\mathrm{Coh}_{\mathrm{fr}}^{\hat{G}}(\hat{G}\sigma)$ |
| Geometric global  | $\mathrm{D}\text{-mod}(\mathrm{Bun}_G)$                 | $\mathrm{IndCoh}(\mathrm{LocSys}_{\hat{G}})$          |

Each row is a **categorification** of the one above.

Each column is connected by duality.

# Three Roles of Geometric Satake



1. **Special case** of LLC (hyperspecial level = unramified)
2. **Source of Hecke operators** ( $\text{IC}_V$  defines  $T_V$ )
3. **Starting point** of the level hierarchy (hyperspecial  $\rightarrow$  Iwahori  $\rightarrow \dots$ )

- **Wild ramification:** No geometric description of LLC for arbitrary depth. Wild inertia destroys semisimplicity at all levels.  
Frenkel–Gross rigid connection, Kaletha’s regular supercuspidals.
- **$p$ -adic coefficients:** Replace  $\overline{\mathbb{Q}}_\ell$  by  $\overline{\mathbb{Z}}_\ell$  or  $\overline{\mathbb{F}}_\ell$ . Geometric Satake holds [MV07, Zhu], but semisimplicity fails  $\Rightarrow$  modular representation theory enters.
- **Fargues–Scholze:** Geometrization of LLC via the Fargues–Fontaine curve — “ $p$ -adic Geometric Langlands.”

Thank you!

Questions?